# Gain-scheduled Feedforward-feedback Control for Reducing Lateral Acceleration and Jerk of Vehicle during Lane Changing Maneuver

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*This study proposes a gain-scheduled feedforward-feedback control method to enhance vehicle lane changing performance by reducing the vehicle's lateral acceleration and jerk. In typical lane changing scenarios at constant longitudinal cruising speeds, excessive lateral acceleration and jerk should be minimized to avoid passenger discomfort. However, conventional path-following feedback controllers react to errors only after they occur, creating a trade-off between reducing lagging behavior and minimizing lateral acceleration and jerk. For example, increasing the lane-keeping control gain reduces the lagging behavior but amplifies a control effort, leading to unnecessarily high lateral acceleration and jerk. The proposed control method combines a gain-scheduled feedforward controller with a feedback controller to overcome these limitations. For a path-following feedback controller, a linear-quadratic-regulator (LQR) controller that uses errors with respect to the reference path as a state vector is used. The vehicle's dynamics model used for the LQR controller is a 2-DOF dual-track vehicle model based on a linear tyre model, which accepts the Ackerman-geometry front-wheel steering angle as input and outputs vehicle's yaw rate and sideslip angle. A Luenberger-PI observer is incorporated into the control loop, as typical vehicle sensors cannot measure derivatives of distance errors and yaw rates relative to the reference trajectory. To minimize the lagging behavior of the feedback controller, a feedforward controller is implemented. First, a steering angle for a time-constant radius trajectory can be calculated. Assuming the trajectory radius is large enough, the steering angle for a time-constant radius trajectory can approximate that for a time-varying radius trajectory. To address the initial offset distance from the reference trajectory, which can increase a control effort and thereby vehicle lateral jerk, a rational function-based and a quadratic function-based gain-schedulers are incorporated into the feedforward controller and feedback controller, respectively. Finally, the total steering angle is calculated as the summation of the steering angles from the feedforward controller and feedback controller. The proposed control method is validated through MATLAB/Simulink simulations with vehicle longitudinal speeds ranging from 60 kph to 120 kph. Results demonstrate that the proposed control strategy effectively reduces vehicle's lateral acceleration and jerk by average values of 36.4 and 79.6%, respectively, and lateral distance error by an average value of 54.2%, compared to a feedback-only controller.*

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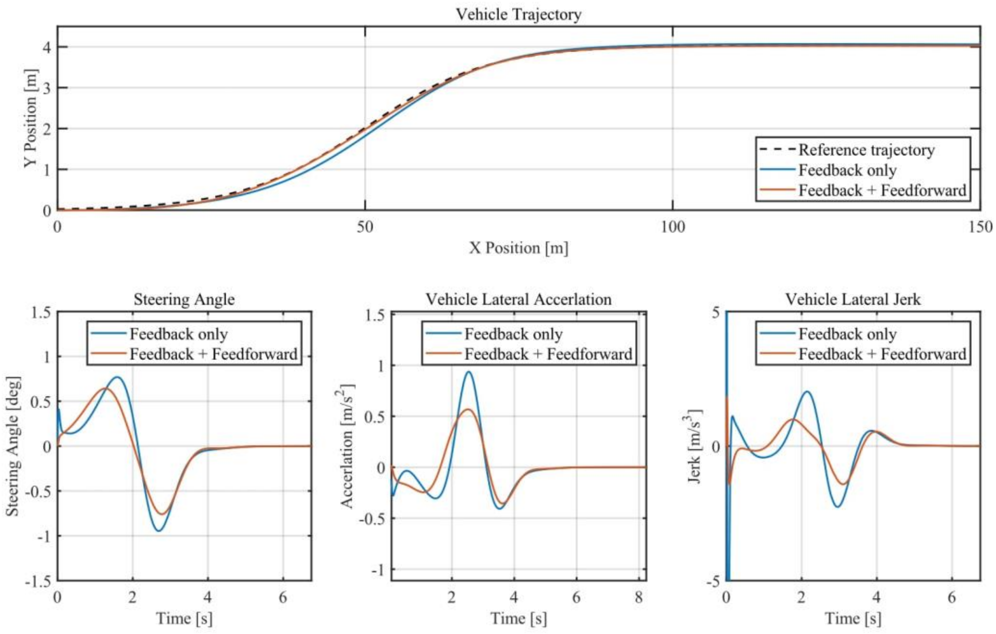


Fig. 1 Comparison between a feedback controller and a gain-scheduled feedforward controller integrated into a feedback loop